

2 A satellite is in geostationary orbit around the Earth. The orbit is circular.

(a) (i) State what is meant by a geostationary orbit.

.....
.....
..... [1]

(ii) State **two** properties of the orbit.

1
.....
2
..... [2]

(b) The orbit has radius R and period T .

(i) Determine an expression, in terms of R and T , for the speed v of the satellite in its orbit.

$v =$ [1]

(ii) Use Newton's law of gravitation to show that the mass M of the Earth is given by

$$M = \frac{4\pi^2 R^3}{GT^2}$$

where G is the gravitational constant.

[2]

- (c) The Earth may be considered to be a sphere of radius 6370 km.
Geostationary satellites orbit at a height of 35 900 km above the surface of the Earth.

Determine a value, to three significant figures, for the mean density of the Earth.

density = kg m^{-3} [4]

[Total: 10]